GANPAT UNIVERSITY
ACHARYA MOTIBHAI PATEL INSTITUTE OF COMPUTER STUDIES (AMPICS)

Student Central Administrative System (SCAS)

System Database Architecture & Relational Schema Specification

Document Section: 4.2 Data Dictionary

1. Specification Overview & Standard Design

This document details the comprehensive Data Dictionary for the **Student Central Administrative System (SCAS)**, designed in accordance with standard relational database documentation specifications. Every entity represents an active table in the MySQL database schema powering authentication, curriculum management, automated timetabling, CIE evaluations, diagnostic unit testing, and fee transaction accounting.

Reference Design Standard (Demo Specification):

Table Name: companies
Description: Stores company information such as company identity, login details, contact information, website, location, and company logo for managing placement-related organizations.
Constraints: company_id (PK), email (UNIQUE), company_name (NOT NULL), email (NOT NULL), password (NOT NULL), created_at (NOT NULL, DEFAULT CURRENT_TIMESTAMP)

Field No.	Field Name	Type	Size	Constraints	Description
1	**company_id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique identifier for each company
2	company_name	VARCHAR	150	NOT NULL	Name of the company
3	email	VARCHAR	100	UNIQUE, NOT NULL	Unique email address used for company login/contact
4	password	VARCHAR	255	NOT NULL	Encrypted/hashed password for company account

Note: All 13 relational database tables below strictly adopt this validated structural schema format.

2. Master Index of SCAS System Tables (13 Entities)

The following master index provides a consolidated register of all 13 core relational entities implemented within the SCAS database engine, indexed by section number, table identifier, primary key, and functional purpose:

Table #	Physical Table Name	Primary Key	Functional Subsystem Role
4.2.1	scas_users	id	Security, RBAC & Synthetic User Credentials
4.2.2	scas_courses	id	Curricular Degrees (MCA, BCA, B.Tech)
4.2.3	scas_batches	id	Academic Cohorts & Graduation Batches
4.2.4	scas_faculty_profiles	id	Faculty Metadata & Department Assignments
4.2.5	scas_student_profiles	id	Student Roll Numbers & Demographics
4.2.6	scas_subjects	id	Academic Curriculum Subjects & Faculty Links
4.2.7	scas_schedules	id	Weekly Lecture Timetable & Room Allocations
4.2.8	scas_attendance	id	Daily Lecture Attendance Logs
4.2.9	scas_marks	id	Continuous Internal Evaluation (CIE) Marks
4.2.10	scas_results	id	Semester Official Grade Sheets & Credits
4.2.11	scas_test_assessments	id	Diagnostic MCQ Engine & Mastery Taxonomy
4.2.12	scas_fee_payments	id	Student Tuition Ledger & Transaction Receipts
4.2.13	scas_leave_applications	id	Digital Leave Workflow & Approval Auditing

Table 4.2.1: scas_users

Table Name: scas_users
Description: Stores system authentication credentials, user identity profiles, role-based access control (RBAC), and account creation timestamps for administrative, faculty, and student users.
Constraints: id (PK), username (UNIQUE, NOT NULL), password_hash (NOT NULL), email (UNIQUE, NOT NULL), role (NOT NULL), name (NOT NULL), created_at (NOT NULL, DEFAULT CURRENT_TIMESTAMP)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique internal synthetic user identifier; primary key.
2	username	VARCHAR	80	UNIQUE, NOT NULL	System login identifier; globally unique across all system roles.
3	password_hash	VARCHAR	255	NOT NULL	Cryptographic password hash generated via Scrypt (Werkzeug security).
4	email	VARCHAR	120	UNIQUE, NOT NULL	Primary contact email address; globally unique; used for notices & resets.
5	role	VARCHAR	20	NOT NULL	Role-Based Access Control enum: 'admin', 'faculty', or 'student'.
6	name	VARCHAR	100	NOT NULL	Full legal name of the user for institutional records & certificates.
7	created_at	DATETIME	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	UTC timestamp recording account creation time.

Relational Integrity & Schema Notes for scas_users:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.2: scas_courses

Table Name: scas_courses
Description: Stores academic degree programs, program identification codes, and comprehensive curriculum syllabus guidelines.
Constraints: id (PK), code (UNIQUE, NOT NULL), name (NOT NULL), description (NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique identifier for the degree program / course.
2	name	VARCHAR	100	NOT NULL	Full academic degree name (e.g., 'Master of Computer Applications').
3	code	VARCHAR	20	UNIQUE, NOT NULL	Short unique institutional course code (e.g., 'MCA', 'BCA', 'B.Tech').
4	description	TEXT	-	NULL	Detailed curriculum overview, program syllabus objectives, and guidelines.

Relational Integrity & Schema Notes for scas_courses:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.3: scas_batches

Table Name: scas_batches
Description: Stores academic cohort batches, section designations, graduation/completion calendar years, and degree program associations.
Constraints: id (PK), course_id (FK, NOT NULL), name (NOT NULL), year (NOT NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique identifier for academic cohort batch.
2	name	VARCHAR	50	NOT NULL	Cohort identifier name (e.g., 'MCA 2024-2026', 'MCA Section A').
3	course_id	INT	11	FK, NOT NULL	Foreign key referencing scas_courses(id) ON DELETE CASCADE.
4	year	INT	11	NOT NULL	Graduation or passing calendar year (e.g., 2026).

Relational Integrity & Schema Notes for scas_batches:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.4: scas_faculty_profiles

Table Name: scas_faculty_profiles
Description: Stores faculty academic metadata, teaching departments, and institutional designations as a 1:1 extension of the user account.
Constraints: id (PK), user_id (FK, UNIQUE, NOT NULL), department (NOT NULL), designation (NOT NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique faculty profile record identifier.
2	user_id	INT	11	FK, UNIQUE, NOT NULL	Foreign key referencing scas_users(id) ON DELETE CASCADE (1:1 extension).
3	department	VARCHAR	100	NOT NULL	Academic teaching department (e.g., 'Department of Computer Science & Applications').
4	designation	VARCHAR	100	NOT NULL	Academic rank/title (e.g., 'Associate Professor', 'Assistant Professor', 'HOD').

Relational Integrity & Schema Notes for scas_faculty_profiles:
- **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
- **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
- **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.5: scas_student_profiles

Table Name: scas_student_profiles
Description: Stores student enrollment profiles, university hall ticket/matriculation numbers, cohort batch links, course allocations, and contact details.
Constraints: id (PK), user_id (FK, UNIQUE, NOT NULL), roll_no (UNIQUE, NOT NULL), batch_id (FK, NOT NULL), course_id (FK, NOT NULL), phone (NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique student profile record identifier.
2	user_id	INT	11	FK, UNIQUE, NOT NULL	Foreign key referencing scas_users(id) ON DELETE CASCADE (1:1 extension).
3	roll_no	VARCHAR	30	UNIQUE, NOT NULL	University matriculation / hall ticket roll number (e.g., 'MCA2401', '24MCA042').
4	batch_id	INT	11	FK, NOT NULL	Foreign key referencing scas_batches(id) ON DELETE CASCADE.
5	course_id	INT	11	FK, NOT NULL	Foreign key referencing scas_courses(id) ON DELETE CASCADE.
6	phone	VARCHAR	20	NULL	Student / emergency guardian contact phone number for SMS and notifications.

Relational Integrity & Schema Notes for scas_student_profiles:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.6: scas_subjects

Table Name: scas_subjects
Description: Stores academic curriculum subjects, subject course codes, degree course associations, and assigned faculty educator references.
Constraints: id (PK), code (UNIQUE, NOT NULL), name (NOT NULL), course_id (FK, NOT NULL), faculty_id (FK, NOT NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique curriculum course subject identifier.
2	name	VARCHAR	100	NOT NULL	Full academic subject title (e.g., 'Advanced Database Systems', 'Cloud Computing').
3	code	VARCHAR	20	UNIQUE, NOT NULL	Official academic subject course code (e.g., 'MCA201', 'CS304').
4	course_id	INT	11	FK, NOT NULL	Foreign key referencing scas_courses(id) ON DELETE CASCADE.
5	faculty_id	INT	11	FK, NOT NULL	Foreign key referencing scas_users(id) ON DELETE CASCADE (Assigned teacher).

Relational Integrity & Schema Notes for scas_subjects:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.7: scas_schedules

Table Name: scas_schedules
Description: Stores weekly timetable lecture schedules, room allocations, days of the week, and commencement/termination time slots.
Constraints: id (PK), subject_id (FK, NOT NULL), batch_id (FK, NOT NULL), day_of_week (NOT NULL), start_time (NOT NULL), end_time (NOT NULL), room (NOT NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique timetable slot identifier.
2	subject_id	INT	11	FK, NOT NULL	Foreign key referencing scas_subjects(id) ON DELETE CASCADE.
3	batch_id	INT	11	FK, NOT NULL	Foreign key referencing scas_batches(id) ON DELETE CASCADE.
4	day_of_week	VARCHAR	20	NOT NULL	Scheduled lecture day: 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday'.
5	start_time	VARCHAR	10	NOT NULL	Lecture commencement time string (e.g., '09:00 AM', '11:30 AM').
6	end_time	VARCHAR	10	NOT NULL	Lecture termination time string (e.g., '10:00 AM', '12:30 PM').
7	room	VARCHAR	20	NOT NULL	Allocated classroom / computer laboratory identifier (e.g., 'Lab 3', 'Hall 201').

Relational Integrity & Schema Notes for scas_schedules:
- **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
- **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
- **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.8: scas_attendance

Table Name: scas_attendance
Description: Stores daily attendance event logs recording student presence or absence status per scheduled subject lecture.
Constraints: id (PK), student_id (FK, NOT NULL), subject_id (FK, NOT NULL), date (NOT NULL), status (NOT NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique attendance log transaction identifier.
2	student_id	INT	11	FK, NOT NULL	Foreign key referencing scas_student_profiles(id) ON DELETE CASCADE.
3	subject_id	INT	11	FK, NOT NULL	Foreign key referencing scas_subjects(id) ON DELETE CASCADE.
4	date	DATE	-	NOT NULL	Calendar date on which attendance was recorded (YYYY-MM-DD).
5	status	VARCHAR	10	NOT NULL	Daily attendance status value: 'Present' or 'Absent'.

Relational Integrity & Schema Notes for scas_attendance:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.9: scas_marks

Table Name: scas_marks
Description: Stores continuous internal evaluations (CIE), examination classifications, scores achieved, and maximum benchmark marks.
Constraints: id (PK), student_id (FK, NOT NULL), subject_id (FK, NOT NULL), exam_type (NOT NULL), marks_obtained (NOT NULL), max_marks (NOT NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique internal evaluation score entry identifier.
2	student_id	INT	11	FK, NOT NULL	Foreign key referencing scas_student_profiles(id) ON DELETE CASCADE.
3	subject_id	INT	11	FK, NOT NULL	Foreign key referencing scas_subjects(id) ON DELETE CASCADE.
4	exam_type	VARCHAR	30	NOT NULL	Examination classification: 'Midterm 1', 'Midterm 2', 'Assignment', 'Final Exam'.
5	marks_obtained	FLOAT	-	NOT NULL	Numeric score achieved by student in the assessment (e.g., 42.5).
6	max_marks	FLOAT	-	NOT NULL	Maximum total marks possible for this examination type (e.g., 50.0 or 100.0).

Relational Integrity & Schema Notes for scas_marks:
• **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
• **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
• **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.10: scas_results

Table Name: scas_results
Description: Stores semester grade sheets, final aggregated scores, benchmark marks, and academic credits earned by students.
Constraints: id (PK), student_id (FK, NOT NULL), subject_id (FK, NOT NULL), semester (NOT NULL), marks_obtained (NOT NULL), max_marks (NOT NULL), credits (NOT NULL, DEFAULT 4)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique semester result sheet entry identifier.
2	student_id	INT	11	FK, NOT NULL	Foreign key referencing scas_student_profiles(id) ON DELETE CASCADE.
3	subject_id	INT	11	FK, NOT NULL	Foreign key referencing scas_subjects(id) ON DELETE CASCADE.
4	semester	INT	11	NOT NULL	Academic semester index (e.g., 1, 2, 3, 4 for MCA program).
5	marks_obtained	FLOAT	-	NOT NULL	Final aggregate score obtained out of 100 in the semester.
6	max_marks	FLOAT	-	NOT NULL	Benchmark maximum aggregate marks (standard 100.0).
7	credits	INT	11	NOT NULL, DEFAULT 4	Academic credits assigned to this course subject (default = 4 credits).

Relational Integrity & Schema Notes for scas_results:
- **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
- **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
- **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.11: scas_test_assessments

Table Name: scas_test_assessments
Description: Stores continuous unit diagnostic evaluations, pre/post MCQ test transactions, raw scores, percentage metrics, and mastery taxonomy.
Constraints: id (PK), student_id (FK, NOT NULL), subject_id (FK, NOT NULL), unit_number (NOT NULL), unit_title (NOT NULL), test_type (NOT NULL), score (NOT NULL), percentage (NOT NULL), total_questions (DEFAULT 25), mastery_level (DEFAULT 'Intermediate'), submitted_at (DEFAULT CURRENT_TIMESTAMP)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique test assessment attempt transaction identifier.
2	student_id	INT	11	FK, NOT NULL	Foreign key referencing scas_student_profiles(id) ON DELETE CASCADE.
3	subject_id	INT	11	FK, NOT NULL	Foreign key referencing scas_subjects(id) ON DELETE CASCADE.
4	unit_number	INT	11	NOT NULL	Curriculum unit sequence number (e.g., Unit 1 through Unit 5).
5	unit_title	VARCHAR	200	NOT NULL	Descriptive topic / chapter heading covered by this assessment.
6	test_type	VARCHAR	20	NOT NULL	Diagnostic evaluation classification: 'pre-test' or 'post-test'.
7	score	FLOAT	-	NOT NULL	Raw marks achieved in test (typically out of 25.0 points).
8	total_questions	INT	11	NOT NULL, DEFAULT 25	Total questions presented in the multiple-choice assessment.
9	percentage	FLOAT	-	NOT NULL	Calculated performance percentage: (score / total_questions) * 100.
10	mastery_level	VARCHAR	50	DEFAULT 'Intermediate'	Categorical taxonomy: 'Beginner', 'Intermediate', or 'Mastered'.
11	submitted_at	DATETIME	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Submission timestamp for speed, telemetry & placement leaderboard.

Relational Integrity & Schema Notes for scas_test_assessments:
- **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
- **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
- **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.12: scas_fee_payments

Table Name: scas_fee_payments
Description: Stores student financial fee ledgers, transaction records, clearance status, institutional receipts, and payment gateway references.
Constraints: id (PK), student_id (FK, NOT NULL), amount (NOT NULL), status (NOT NULL), payment_date (NOT NULL, DEFAULT CURRENT_TIMESTAMP), receipt_no (UNIQUE, NULL), transaction_id (UNIQUE, NULL)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique fee transaction ledger identifier.
2	student_id	INT	11	FK, NOT NULL	Foreign key referencing scas_student_profiles(id) ON DELETE CASCADE.
3	amount	FLOAT	-	NOT NULL	Transaction amount in institutional currency (e.g., 45000.00).
4	payment_date	DATETIME	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Timestamp when payment was confirmed or recorded in ledger.
5	status	VARCHAR	20	NOT NULL	Fee clearance state: 'Paid', 'Pending', or 'Overdue'.
6	receipt_no	VARCHAR	50	UNIQUE, NULL	Unique institutional receipt sequence number (e.g., 'REC-2026-0814').
7	transaction_id	VARCHAR	50	UNIQUE, NULL	External banking / UPI reference transaction code (e.g., 'TXN938201948').

Relational Integrity & Schema Notes for scas_fee_payments:
- **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
- **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
- **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.

Table 4.2.13: scas_leave_applications

Table Name: scas_leave_applications
Description: Stores institutional leave applications, leave categories, absence date spans, applicant justifications, and administrative approval workflow states.
Constraints: id (PK), user_id (FK, NOT NULL), leave_type (NOT NULL), start_date (NOT NULL), end_date (NOT NULL), reason (NOT NULL), status (NOT NULL, DEFAULT 'Pending'), applied_at (NOT NULL, DEFAULT CURRENT_TIMESTAMP)

Field No.	Field Name	Type	Size	Constraints	Description
1	**id**	INT	11	PK, AUTO_INCREMENT, NOT NULL	Unique leave application identifier.
2	user_id	INT	11	FK, NOT NULL	Foreign key referencing scas_users(id) ON DELETE CASCADE (Applicant).
3	leave_type	VARCHAR	50	NOT NULL	Leave classification: 'Sick Leave', 'Casual Leave', 'Duty Leave', 'Maternity'.
4	start_date	DATE	-	NOT NULL	Inclusive start date of proposed absence (YYYY-MM-DD).
5	end_date	DATE	-	NOT NULL	Inclusive completion date of proposed absence (YYYY-MM-DD).
6	reason	TEXT	-	NOT NULL	Applicant's textual justification, medical note details, or on-duty event brief.
7	status	VARCHAR	20	NOT NULL, DEFAULT 'Pending'	Workflow lifecycle approval status: 'Pending', 'Approved', 'Rejected'.
8	applied_at	DATETIME	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Submission timestamp for review SLA tracking.

Relational Integrity & Schema Notes for scas_leave_applications:
- **Primary Key:** id (Auto-incrementing integer identifier guaranteeing unique row resolution).
- **Referential Action:** Foreign key relationships utilize ON DELETE CASCADE to preserve referential consistency across parent entities.
- **Storage Engine:** InnoDB | **Character Set:** utf8mb4 | **Collation:** utf8mb4_unicode_ci.